

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Classified Practice

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Classified Practice

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Classified Practice

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Choose the correct option.

A $4 + 8 + 12$

B $4 + 8 + 16$

C $8 + 12 + 16$

D $4 + 12 + 20$

Classified Practice

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Choose the correct option.

A $2 + 4 + 8 + 16$

B $1 + 2 + 4 + 8$

C $2 + 6 + 18 + 54$

D $4 + 8 + 16 + 32$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Choose the correct option.

A $4 + 24 + 124$

B $5 + 25 + 125$

C $4 + 25 + 124$

D $1 + 4 + 24$

Classified Practice

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Choose the correct option.

A $5 + 6 + 7 + 8 + 9$

B $4 + 5 + 6 + 7 + 8$

C $1 + 2 + 3 + 4 + 5$

D $6 + 7 + 8 + 9 + 10$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Choose the correct option.

A $7 + 9 + 11 + 13$

B $5 + 7 + 9 + 11$

C $4 + 7 + 10 + 13$

D $8 + 10 + 12 + 14$

Classified Practice

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Choose the correct option.

- A $3 + 5 + 9$
- B $2 + 4 + 8$
- C $1 + 3 + 5$
- D $4 + 6 + 10$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Choose the correct option.

A $4 + 10 + 18$

B $3 + 8 + 15$

C $1 + 4 + 9$

D $6 + 12 + 18$

Classified Practice

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $5 + 10 + 17 + 26$

B $2 + 5 + 10 + 17$

C $4 + 9 + 16 + 25$

D $6 + 11 + 18 + 27$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Classified Practice

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Classified Practice

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Classified Practice

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Choose the correct option.

A $3 + 5 + 7 + 9$

B $2 + 4 + 6 + 8$

C $3 + 6 + 9 + 12$

D $1 + 3 + 5 + 7$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Choose the correct option.

A $5 + 8 + 11 + 14$

B $2 + 5 + 8 + 11$

C $5 + 9 + 13 + 17$

D $4 + 7 + 10 + 13$

Classified Practice

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Choose the correct option.

A $-1 + 5 + 15 + 29$

B $-1 + 1 + 5 + 13$

C $2 + 5 + 10 + 17$

D $-3 + 5 + 13 + 29$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Choose the correct option.

A $4 + 13 + 28 + 49$

B $3 + 12 + 27 + 48$

C $4 + 10 + 19 + 28$

D $1 + 4 + 9 + 16$

Classified Practice

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Choose the correct option.

A $0 + 2 + 6 + 12$

B $1 + 4 + 9 + 16$

C $0 + 1 + 4 + 9$

D $2 + 6 + 12 + 20$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Choose the correct option.

A $10 + 15 + 20$

B $5 + 10 + 15$

C $2 + 4 + 6$

D $15 + 20 + 25$

Classified Practice

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Choose the correct option.

A $3 + 3 + 3 + 3$

B $3 + 6 + 9 + 12$

C $1 + 1 + 1 + 1$

D $4 + 4 + 4 + 4$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Choose the correct option.

A $6 + 11 + 16 + 21$

B $5 + 10 + 15 + 20$

C $1 + 6 + 11 + 16$

D $7 + 12 + 17 + 22$

Classified Practice

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Classified Practice

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Classified Practice

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Classified Practice

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Classified Practice

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Classified Practice

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Choose the correct option.

A $3 + 6 + 11$

B $2 + 5 + 10$

C $3 + 5 + 9$

D $1 + 4 + 9$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Choose the correct option.

A $7 + 7 + 7 + 7 + 7$

B $7 + 14 + 21 + 28 + 35$

C $7 + 7 + 7 + 7$

D $5 + 5 + 5 + 5 + 5$

Classified Practice

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Choose the correct option.

A $9 + 16 + 25 + 36$

B $1 + 4 + 9 + 16$

C $9 + 25 + 49 + 81$

D $3 + 4 + 5 + 6$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Choose the correct option.

A $6 + 36 + 216$

B $1 + 6 + 36$

C $6 + 12 + 18$

D $36 + 72 + 108$

Classified Practice

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Choose the correct option.

A $1 + 5 + 9 + 13 + 17$

B $4 + 8 + 12 + 16 + 20$

C $1 + 4 + 7 + 10 + 13$

D $-1 + 3 + 7 + 11 + 15$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Choose the correct option.

A $-2 + 1 + 4 + 7$

B $-2 + 0 + 2 + 4$

C $3 + 6 + 9 + 12$

D $-5 - 2 + 1 + 4$

Classified Practice

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Choose the correct option.

A $2 + 14 + 62$

B $4 + 16 + 64$

C $2 + 12 + 60$

D $-2 + 2 + 14$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Choose the correct option.

A $1 + 6 + 15$

B $2 + 7 + 18$

C $1 + 3 + 5$

D $0 + 6 + 12$

Classified Practice

Summation

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ using the summation symbol Σ .

Q046 Written**Express $5 + 8 + 11 + \dots + (3n + 2)$ using Σ .**

Classified Practice

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Q048 Written**Express $1 + 3 + 5 + \dots + (2n - 1)$ using Σ .**

Classified Practice

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .

Q050 Written**Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**

Classified Practice

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \dots + (3n - 1)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (3k - 1)$
 - B $\sum_{k=0}^n (3k - 1)$
 - C $\sum_{k=1}^n (3k + 1)$
 - D $\sum_{k=1}^n (2k - 1)$
-
-
-

Classified Practice

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=1}^{10} (4k + 3)$

B $\sum_{k=1}^9 (4k + 3)$

C $\sum_{k=0}^{10} (4k + 3)$

D $\sum_{k=1}^{10} (4k - 3)$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n (4k + 2)$
- B $\sum_{k=0}^n (4k + 2)$
- C $\sum_{k=1}^n (4k - 2)$
- D $\sum_{k=1}^n (3k + 2)$

Classified Practice

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^{20} (5k + 3)$

B $\sum_{k=1}^{19} (5k + 3)$

C $\sum_{k=0}^{20} (5k + 3)$

D $\sum_{k=1}^{20} (5k - 3)$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=11}^{19} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=11}^{19} k$

D $\sum_{k=12}^{19} k^2$

Classified Practice

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=2}^{10} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=2}^{10} k$

D $\sum_{k=3}^{10} k^2$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=0}^{n-1} 3^k$

B $\sum_{k=1}^n 3^k$

C $\sum_{k=1}^{n-1} 3^k$

D $\sum_{k=0}^n 3^k$

Classified Practice

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \cdots + (5n - 3)$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n (5k - 3)$

B $\sum_{k=0}^n (5k - 3)$

C $\sum_{k=1}^n (5k + 3)$

D $\sum_{k=1}^n (4k - 3)$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Classified Practice

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \dots + 13^2$ using Σ .

Q062 Written**Express $1 + 7 + 25 + 79 + \dots + (3^n - 2)$ using Σ .**

Classified Practice

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Q064 Written**Express $7^2 + 8^2 + \dots + 21^2$ using Σ .**

Classified Practice

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ using Σ .

Q066 Written**Express $10^2 + 11^2 + \dots + 20^2$ using Σ .**

Classified Practice

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=4}^{12} k^2$

B $\sum_{k=1}^9 k^2$

C $\sum_{k=4}^{12} k$

D $\sum_{k=5}^{12} k^2$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 3^k$

B $\sum_{k=0}^n 3^k$

C $\sum_{k=1}^n 3^{k-1}$

D $\sum_{k=1}^n k^3$

Classified Practice

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=1}^n 2^k$
- B $\sum_{k=0}^n 2^k$
- C $\sum_{k=1}^n 2^{k-1}$
- D $\sum_{k=1}^n k^2$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :

Choose the correct option.

- A $\sum_{k=0}^{n-1} 2^k$
- B $\sum_{k=1}^n 2^k$
- C $\sum_{k=1}^{n-1} 2^k$
- D $\sum_{k=0}^n 2^k$

Classified Practice

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=1}^n 7k$

B $\sum_{k=0}^n 7k$

C $\sum_{k=1}^n (7k + 1)$

D $\sum_{k=1}^n k^7$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Choose the correct option.**

A $\sum_{k=2}^n 5k$

B $\sum_{k=1}^n 5k$

C $\sum_{k=2}^n (5k + 1)$

D $\sum_{k=0}^n 5k$

Classified Practice

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Choose the correct option.

A $\sum_{k=6}^{16} k^2$

B $\sum_{k=1}^{11} k^2$

C $\sum_{k=6}^{16} k$

D $\sum_{k=7}^{16} k^2$

Q074 Written**Express $2 + 5 + 8 + \cdots + 29$ using Σ .**

Classified Practice

Summation

Q075 Written

Express $8 + 16 + 24 + \cdots + 56$ using Σ .

Q076 Written**Express $7 + 13 + 19 + \dots + 73$ using Σ .**

Classified Practice

Summation

Q077 Written

Express $4 + 9 + 14 + \cdots + 119$ using Σ .

Classified Practice

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Classified Practice

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Q080 Written**Find** $\sum_{k=1}^n 3.$

Classified Practice

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Classified Practice

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Choose the correct option.

A $9n$

B $n + 9$

C $9n^2$

D $\frac{9n(n+1)}{2}$

Classified Practice

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Choose the correct option.

A $-2n$

B $2n$

C $-n - 2$

D $-n^2$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Choose the correct option.

A 28

B 11

C 16

D 49

Classified Practice

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Choose the correct option.

A -60

B 60

C -17

D -144

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Choose the correct option.

A $\frac{n(n+1)}{2}$

B $n(n+1)$

C $\frac{n(n-1)}{2}$

D n^2

Classified Practice

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Choose the correct option.

A $2n(n+1)$

B $4n(n+1)$

C $2n(n-1)$

D $4n$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Choose the correct option.

A 55

B 45

C 100

D 110

Classified Practice

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Choose the correct option.

A 108

B 72

C 96

D 120

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Choose the correct option.

A 105

B 75

C 90

D 120

Classified Practice

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Choose the correct option.

- A 210
- B 190
- C 200
- D 420

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Choose the correct option.

A $\frac{n(n+1)(2n+1)}{6}$

B $\frac{n(n+1)^6}{2}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n^2(n+1)^2}{4}$

Classified Practice

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Choose the correct option.

A $n(n+1)(2n+1)$

B $3n(n+1)$

C $n(n+1)(2n-1)$

D $6n^2$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Choose the correct option.

A 55

B 25

C 30

D 50

Classified Practice

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Choose the correct option.

A 60

B 30

C 40

D 80

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Choose the correct option.

A 56

B 36

C 48

D 72

Classified Practice

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Choose the correct option.

A 140

B 49

C 91

D 196

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Classified Practice

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Classified Practice

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Classified Practice

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Choose the correct option.

A $\frac{n(3n+1)}{2}$

B $\frac{n(3n-1)}{2}$

C $n(3n + 1)$

D $\frac{3n(n+1)}{2}$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Choose the correct option.

A $n(n + 6)$

B $n(n + 5)$

C $n(n + 1) + 5$

D $2n(n + 6)$

Classified Practice

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Choose the correct option.

A $4n^2$

B $4n(n + 1)$

C $8n^2$

D $4n(n - 1)$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Choose the correct option.

A $\frac{n(5n+11)}{2}$

B $\frac{n(5n+3)}{2}$

C $\frac{n(5n+5)}{2}$

D $n(5n + 11)$

Classified Practice

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)}{6}$

B $\frac{n(n+1)(2n+1)}{6}$

C $\frac{n(n+1)(n+2)}{6}$

D $n(n+1)(n+2)$

Classified Practice

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Choose the correct option.

A $\frac{n(n+1)(4n-1)}{6}$

B $\frac{n(n+1)(4n+1)}{6}$

C $\frac{n(n+1)(2n-1)}{6}$

D $\frac{n(n-1)(4n-1)}{6}$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Choose the correct option.

A $\frac{n(2n^2+3n-5)}{6}$

B $\frac{n(2n^2+3n-1)}{6}$

C $\frac{n(2n^2-3n+1)}{6}$

D $\frac{n(n+1)(2n+1)}{6}$

Classified Practice

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Choose the correct option.

A $\frac{n(n+1)(2n+3)}{2}$

B $\frac{n(n+1)(2n+1)}{2}$

C $n(n+1)(2n+3)$

D $\frac{n(n-1)(2n+3)}{2}$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Choose the correct option.

A $\frac{n(8n^2+3n+7)}{6}$

B $\frac{n(8n^2+9n+7)}{6}$

C $\frac{n(4n^2+3n+2)}{6}$

D $\frac{n(8n^2-3n+7)}{6}$

Classified Practice

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Classified Practice

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Q118 Written

Find $\sum_{k=1}^n k(k-2)$.

Classified Practice

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n + 3)$.

Classified Practice

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n + 1)(n + 4)$.

Classified Practice

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Classified Practice

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Classified Practice

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Classified Practice

Summation

Q128 **Challenge**

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Classified Practice

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Classified Practice

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Classified Practice

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:

Choose the correct option.

A $\frac{n(n+1)(n+2)(3n+1)}{12}$

B $\frac{n(n+1)(n+2)(3n-1)}{12}$

C $\frac{n(n+1)(3n+1)}{12}$

D $\frac{n(n+1)(n+2)(3n+2)}{6}$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2+5n-4)}{12}$
- C $\frac{n(n+1)(3n^2-5n+4)}{12}$
- D $\frac{n(n+1)(3n^2-4n-5)}{12}$

Classified Practice

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+11n+4)}{12}$

B $\frac{n(n+1)(3n^2+11n-4)}{12}$

C $\frac{n(n+1)(3n^2+9n+4)}{12}$

D $\frac{n(n+1)(3n^2+4n+11)}{12}$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2-3n-2)}{4}$

B $\frac{n(n+1)(n^2+3n-2)}{4}$

C $\frac{n(n+1)(n^2-3n+2)}{4}$

D $\frac{n(n+1)(n^2-2n-3)}{4}$

Classified Practice

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+5n+2)}{4}$

B $\frac{n(n+1)(n^2+5n-2)}{4}$

C $\frac{n(n+1)(n^2+3n+2)}{4}$

D $\frac{n(n+1)(n^2+2n+5)}{4}$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-13n-8)}{12}$

B $\frac{n(n+1)(3n^2-13n+8)}{12}$

C $\frac{n(n+1)(3n^2-11n-8)}{12}$

D $\frac{n(n+1)(3n^2-8n-13)}{12}$

Classified Practice

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+19n+8)}{12}$

B $\frac{n(n+1)(3n^2+19n-8)}{12}$

C $\frac{n(n+1)(3n^2+17n+8)}{12}$

D $\frac{n(n+1)(3n^2+8n+19)}{12}$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Choose the correct option.

- A $\frac{n(n+1)(3n^2-5n-4)}{12}$
- B $\frac{n(n+1)(3n^2-5n+4)}{12}$
- C $\frac{n(n+1)(3n^2-4n-5)}{12}$
- D $\frac{n(n+1)(3n^2-n-4)}{12}$

Classified Practice

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Classified Practice

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

A $\frac{n(n+1)(n-1)(n+2)}{4}$

B $\frac{n(n+1)(n+1)(n+2)}{4}$

C $\frac{n(n-1)(n+2)}{4}$

D $\frac{n(n+1)(n-2)(n+3)}{4}$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n+4)}{4}$

B $\frac{n(n+1)(n^2+n-4)}{4}$

C $\frac{n(n+1)(n^2+2n+4)}{4}$

D $\frac{n(n+1)(n^2+n+2)}{4}$

Classified Practice

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+6)}{4}$
- B $\frac{n(n+1)(n^2+n-6)}{4}$
- C $\frac{n(n+1)(n^2+3n+6)}{4}$
- D $\frac{n(n+1)(n^2+n+3)}{4}$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-4)}{4}$

B $\frac{n(n+1)(n^2+n+4)}{4}$

C $\frac{n(n+1)(n^2-2n-4)}{4}$

D $\frac{n(n+1)(n^2+n-2)}{4}$

Classified Practice

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+8)}{4}$
- B $\frac{n(n+1)(n^2+n-8)}{4}$
- C $\frac{n(n+1)(n^2+4n+8)}{4}$
- D $\frac{n(n+1)(n^2+n+4)}{4}$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n-6)}{4}$
- B $\frac{n(n+1)(n^2+n+6)}{4}$
- C $\frac{n(n+1)(n^2-3n-6)}{4}$
- D $\frac{n(n+1)(n^2+n-3)}{4}$
-
-
-

Classified Practice

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+23n+10)}{12}$

B $\frac{n(n+1)(3n^2+23n-10)}{12}$

C $\frac{n(n+1)(3n^2+21n+10)}{12}$

D $\frac{n(n+1)(3n^2+10n+23)}{12}$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-17n-10)}{12}$

B $\frac{n(n+1)(3n^2-17n+10)}{12}$

C $\frac{n(n+1)(3n^2-15n-10)}{12}$

D $\frac{n(n+1)(3n^2-10n-17)}{12}$

Classified Practice

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n+10)}{4}$

B $\frac{n(n+1)(n^2+n-10)}{4}$

C $\frac{n(n+1)(n^2+5n+10)}{4}$

D $\frac{n(n+1)(n^2+n+5)}{4}$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-10)}{4}$

B $\frac{n(n+1)(n^2+n+10)}{4}$

C $\frac{n(n+1)(n^2-5n-10)}{4}$

D $\frac{n(n+1)(n^2+n-5)}{4}$

Classified Practice

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+27n+12)}{12}$

B $\frac{n(n+1)(3n^2+27n-12)}{12}$

C $\frac{n(n+1)(3n^2+25n+12)}{12}$

D $\frac{n(n+1)(3n^2+12n+27)}{12}$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

Choose the correct option.

- A $\frac{n(n+1)(n^2+n+12)}{4}$
- B $\frac{n(n+1)(n^2+n-12)}{4}$
- C $\frac{n(n+1)(n^2+6n+12)}{4}$
- D $\frac{n(n+1)(n^2+n+6)}{4}$

Classified Practice

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-6)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2-25n-12)}{12}$

B $\frac{n(n+1)(3n^2-25n+12)}{12}$

C $\frac{n(n+1)(3n^2-23n-12)}{12}$

D $\frac{n(n+1)(3n^2-12n-25)}{12}$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:

Choose the correct option.

A $\frac{n(n+1)(n^2+n-12)}{4}$

B $\frac{n(n+1)(n^2+n+12)}{4}$

C $\frac{n(n+1)(n^2-6n-12)}{4}$

D $\frac{n(n+1)(n^2+n-6)}{4}$

Classified Practice

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+31n+14)}{12}$

B $\frac{n(n+1)(3n^2+31n-14)}{12}$

C $\frac{n(n+1)(3n^2+29n+14)}{12}$

D $\frac{n(n+1)(3n^2+14n+31)}{12}$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Choose the correct option.

A $\frac{n(n+1)(3n^2+35n+16)}{12}$

B $\frac{n(n+1)(3n^2+35n-16)}{12}$

C $\frac{n(n+1)(3n^2+33n+16)}{12}$

D $\frac{n(n+1)(3n^2+16n+35)}{12}$

Classified Practice

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Classified Practice

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Classified Practice

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Classified Practice

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Classified Practice

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Classified Practice

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Classified Practice

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Classified Practice

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Classified Practice

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Classified Practice

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Classified Practice

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}$.

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Classified Practice

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Choose the correct option.

A $\frac{n}{n+1}$

B $\frac{1}{n+1}$

C $\frac{2(n+1)}{n}$

D $\frac{n+1}{n}$

Classified Practice

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Choose the correct option.

A $\frac{n}{2(n+2)}$

B $\frac{1}{2(n+2)}$

C $\frac{n}{n+2}$

D $\frac{n+1}{2(n+2)}$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Choose the correct option.

A $\frac{n}{3(n+3)}$

B $\frac{n}{n+3}$

C $\frac{1}{3(n+3)}$

D $\frac{n+1}{3(n+3)}$

Classified Practice

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Choose the correct option.

A $\frac{n}{4(n+4)}$

B $\frac{n}{3(n+4)}$

C $\frac{n+1}{4(n+4)}$

D $\frac{1}{4(n+4)}$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Choose the correct option.

A $\frac{n}{6(n+6)}$

B $\frac{n}{5(n+6)}$

C $\frac{n+1}{6(n+6)}$

D $\frac{1}{6(n+6)}$

Classified Practice

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Choose the correct option.

A $\frac{n}{7(n+7)}$

B $\frac{n}{6(n+7)}$

C $\frac{n+1}{7(n+7)}$

D $\frac{1}{7(n+7)}$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Choose the correct option.

A $\frac{n}{9(n+9)}$

B $\frac{n}{8(n+9)}$

C $\frac{n+1}{9(n+9)}$

D $\frac{1}{9(n+9)}$

Classified Practice

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Choose the correct option.

A $\frac{n}{10(n+10)}$

B $\frac{n}{9(n+10)}$

C $\frac{n+1}{10(n+10)}$

D $\frac{1}{10(n+10)}$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Choose the correct option.

A $\frac{n}{11(n+11)}$

B $\frac{n}{10(n+11)}$

C $\frac{n+1}{11(n+11)}$

D $\frac{1}{11(n+11)}$

Classified Practice

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Classified Practice

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}$.

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}$.

Classified Practice

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Choose the correct option.

- A $\frac{n}{2n+1}$
B $\frac{n}{2(2n+1)}$
C $\frac{1}{2n+1}$
D $\frac{n+1}{2n+1}$

Classified Practice

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Choose the correct option.

- A $\frac{n}{3n+1}$
- B $\frac{n}{3(3n+1)}$
- C $\frac{1}{3n+1}$
- D $\frac{n+1}{3n+1}$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Choose the correct option.

A $\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$

B $\frac{1}{6} \left(1 - \frac{1}{3n+4} \right)$

C $\frac{1}{2(3n+4)}$

D $\frac{n}{2(3n+4)}$

Classified Practice

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Choose the correct option.

A $\frac{n}{2(4n+2)}$

B $\frac{n}{4(4n+2)}$

C $\frac{1}{4n+2}$

D $\frac{n+1}{2(4n+2)}$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Choose the correct option.

A $\frac{n}{3(5n+3)}$

B $\frac{n}{5(5n+3)}$

C $\frac{n+1}{3(5n+3)}$

D $\frac{1}{5n+3}$

Classified Practice

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Choose the correct option.

- A $\frac{n}{4n+1}$
- B $\frac{n}{4(4n+1)}$
- C $\frac{1}{4n+1}$
- D $\frac{n+1}{4n+1}$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Choose the correct option.

A $\frac{n}{2(5n+2)}$

B $\frac{n}{5(5n+2)}$

C $\frac{n+1}{2(5n+2)}$

D $\frac{1}{5n+2}$

Classified Practice

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Choose the correct option.

A $\frac{n}{4(6n+4)}$

B $\frac{n}{6(6n+4)}$

C $\frac{n+1}{4(6n+4)}$

D $\frac{1}{6n+4}$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Classified Practice

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}.$

Classified Practice

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Classified Practice

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n} + \sqrt{n+1}}$.

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Classified Practice

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Classified Practice

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}.$

Classified Practice

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Classified Practice

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Q214 Written**Find the sum** $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Classified Practice

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n$

B $-(3^n n)$

C $(3^n n) + 1$

D $(3^n n) - 1$

Classified Practice

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Choose the correct option.

A $\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}$

B $-\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right)$

C $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) + 1$

D $\left(\frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n+5}}{9}\right) - 1$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Choose the correct option.

A $2^n n + 3 \cdot 2^n - 3$

B $-(2^n n + 3 \cdot 2^n - 3)$

C $(2^n n + 3 \cdot 2^n - 3) + 1$

D $(2^n n + 3 \cdot 2^n - 3) - 1$

Classified Practice

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \cdots + (n + 1) \times 3^{n-1}$.

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+1) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n + 3^n - 1}{4}$

B $-\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right)$

C $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n + 3^n - 1}{4}\right) - 1$

Classified Practice

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$

B $-\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right)$

C $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) + 1$

D $\left(\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}\right) - 1$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 3) \times (3)^{k-1}$:

Choose the correct option.

A $3^n n - 2 \cdot 3^n + 2$

B $-(3^n n - 2 \cdot 3^n + 2)$

C $(3^n n - 2 \cdot 3^n + 2) + 1$

D $(3^n n - 2 \cdot 3^n + 2) - 1$

Classified Practice

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Choose the correct option.

- A $-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$
- B $-\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right)$
- C $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) + 1$
- D $\left(-\frac{6(-2)^n n + 5(-2)^n - 5}{9}\right) - 1$

Classified Practice

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Choose the correct option.

- A $-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$
- B $-\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right)$
- C $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) + 1$
- D $\left(-\frac{4(-3)^n n + 9(-3)^n - 9}{16}\right) - 1$

Classified Practice

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \dots + (2n - 1) \times 3^n$.

Classified Practice

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Choose the correct option.

A $\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$

B $-\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right)$

C $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) + 1$

D $\left(\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}\right) - 1$

Classified Practice

Summation

Q228 **Challenge**

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Classified Practice

Summation

Q229 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the k -th term.

Q230 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the k -th term.

Classified Practice

Summation

Q231 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the k -th term.

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the k -th term.

Classified Practice

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :

Choose the correct option.

A $k(k + 1)$

B $k(k + 3)$

C $2k^2$

D $k(k + 2)$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

Choose the correct option.

A $\frac{3k(k+1)}{2}$

B $\frac{3k(k+3)}{2}$

C $3k^2$

D $\frac{3k(k+2)}{2}$

Classified Practice

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, \dots$, what is a_k :

Choose the correct option.

A $k(k + 3)$

B $k(k + 5)$

C $2k(k + 1)$

D $k(k + 4)$

Q236 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, \dots$, what is a_k :

Choose the correct option.

A $2k^2$

B $2k(k + 2)$

C $2k(2k - 1)$

D $2k(k + 1)$

Classified Practice

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence $5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(3k+7)}{2}$

B $\frac{k(3k+13)}{2}$

C $k(3k+2)$

D $\frac{k(3k+10)}{2}$

Q238 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, \dots$, what is a_k :

Choose the correct option.

A $\frac{k(5k+1)}{2}$

B $\frac{k(5k+11)}{2}$

C $k(5k - 2)$

D $\frac{k(5k+6)}{2}$

Classified Practice

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence $6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, \dots$, what is a_k :

Choose the correct option.

A $k(k + 5)$

B $k(k + 7)$

C $2k(k + 2)$

D $k(k + 6)$

Q240 MCQ

Select the correct option: For the cumulative sequence $4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, \dots$, what is a_k :

Choose the correct option.

A $2k(k + 1)$

B $2k(k + 3)$

C $4k^2$

D $2k(k + 2)$

Classified Practice

Summation

Q241 Written

For the sequence $3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, \dots$, find the sum from the first term to the n -th term.

Q242 Written

For the sequence $1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, find the sum from the first term to the n -th term.

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, find the sum from the first term to the n -th term.

Classified Practice

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Choose the correct option.

A 40

B 46

C 34

D 52

Classified Practice

Summation

Q246 **Challenge**

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Classified Practice

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Classified Practice

Summation

Q248 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Classified Practice

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Classified Practice

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Classified Practice

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Choose the correct option.

A $2n + 3$

B $2n - 1$

C $2(n + 2)$

D $a_n = 2n + 1$

Classified Practice

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Choose the correct option.

A $4n + 1$

B $2(2n + 3)$

C $a_n = 2(2n + 1)$

D $4n + 5$

Classified Practice

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Choose the correct option.

A $6n$

B $a_n = 6(n - 1)$

C $6n - 5$

D $2(3n - 4)$

Classified Practice

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Choose the correct option.

A $a_n = 2(4n - 3)$

B $4(2n - 1)$

C $8n - 7$

D $2(4n + 1)$

Classified Practice

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Choose the correct option.

A $10n + 3$

B $2(5n - 1)$

C $10(n + 1)$

D $a_n = 10n$

Classified Practice

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Choose the correct option.

A $4n - 1$

B $4n + 5$

C $a_n = 4n + 1$

D $4n + 3$

Classified Practice

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Choose the correct option.

A $3(2n - 1)$

B $2(3n + 2)$

C $a_n = 2(3n - 1)$

D $6n + 1$

Classified Practice

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Choose the correct option.

A $3n^2 + n + 1$

B $3n^2 + 7n + 6$

C $a_n = 3n^2 + n + 2$

D $3n^2 + n + 3$

Classified Practice

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Choose the correct option.

A $2(n+1)(3n+1)$

B $a_n = 2n(3n-2)$

C $2(3n^2 - 2n + 1)$

D $2(n-1)(3n+1)$

Classified Practice

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Choose the correct option.

A $a_n = 9n^2 - 13n + 9$

B $9n^2 - 13n + 12$

C $9n^2 - 13n + 8$

D $9n^2 + 5n + 5$

Classified Practice

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Choose the correct option.

A $3n^2 - 5n + 5$

B $(n - 1)(3n - 2)$

C $3n^2 + n + 2$

D $a_n = 3n^2 - 5n + 4$

Classified Practice

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Choose the correct option.

A $2(3n^2 + 1)$

B $6n^2 - 1$

C $6(n+1)^2$

D $a_n = 6n^2$

Classified Practice

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Choose the correct option.

A $3(n+1)(n+2)$

B $a_n = 3n(n+1)$

C $3n^2 + 3n + 1$

D $3n^2 + 3n - 2$

Classified Practice

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Classified Practice

Summation

Q265 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Classified Practice

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Classified Practice

Summation

Q267 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Classified Practice

Summation

Q268 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Classified Practice

Summation

Q269 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Classified Practice

Summation

Q270 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Classified Practice

Summation

Q271 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Classified Practice

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Classified Practice

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Choose the correct option.**

A -13

B 13

C 14

D 12

Classified Practice

Summation

Q274 **Challenge**

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Classified Practice

Summation

Quiz MCQ 1 [Concept Quiz](#) · [MCQ](#)

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Choose the correct option.

- A $2 + 5 + 8 + 11$
- B $3 + 6 + 9 + 12$
- C $1 + 4 + 7 + 10$
- D $2 + 6 + 10 + 14$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Choose the correct option.

- A $\frac{n(n+1)(n-1)(n+2)}{4}$
- B $\frac{n(n+1)(n+1)(n+2)}{4}$
- C $\frac{n(n-1)(n+2)}{4}$
- D $\frac{n(n+1)(n-2)(n+3)}{4}$

Classified Practice

Summation

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Choose the correct option.

A $(n-1) \times 2^n + 1$

B $-((n-1) \times 2^n + 1)$

C $((n-1) \times 2^n + 1) + 1$

D $((n-1) \times 2^n + 1) - 1$

Classified Practice

Summation

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Choose the correct option.

A 30

B -32

C 32

D 34

Classified Practice

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k+5)$.

Classified Practice

Summation

Quiz WRITTEN 7 Concept Quiz · Written**Find the sum** $\sum_{k=1}^n (4k - 1) \times 3^k$.

Classified Practice

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$